

Complete E-Commerce Checkout & Place Order API Documentation

This document contains the complete specification for integrating the Checkout Preview and Place Order APIs in the Flutter application.

SECTION 1: Checkout Preview API

1. Endpoint Details

Property	Value
Endpoint	/ogaglow/orders/checkout/preview
HTTP Method	POST
Headers	Content-Type: application/json Accept: application/json
Description	Calculates totals, shipping costs, discounts, and validates coupons before order placement.

2. Request Body (Payload)

```
{
  "items": [
    {
      "product": "6a0d5da172241d4c9fafa20c",
      "quantity": 2
    }
  ],
  "couponCode": "SUMMER40"
}
```

3. Response Structure (200 OK)

```
{
  "items": [
    {
      "product": "6a0d5da172241d4c9fafa20c",
      "name": "Hair Oil",
      "image": null,
      "quantity": 2,
      "price": 1320,
      "originalPrice": 2200,
      "shippingPrice": 250,
      "discountAmount": 1760,
      "subtotal": 2640,
      "discount": {
        "type": "percentage",
        "value": 40
      }
    }
  ],
  "priceBreakdown": {
    "itemsPrice": 2640,
    "productDiscount": 1760,
    "shippingPrice": 250,
    "discountAmount": 0,
    "couponDiscount": 0,
    "appliedCouponCode": null,
    "totalPrice": 2890
  },
  "coupon": null,
  "couponError": null
}
```

SECTION 2: Place Order API

1. Endpoint Details

Property	Value
Endpoint	/ogaglow/orders/placeOrder
HTTP Method	POST
Headers	Content-Type: application/json Accept: application/json
Description	Creates a new order, books Leopards Courier tracking, updates inventory, and triggers notification emails.

2. Request Body (Payload)

```
{
  "userId": null,
  "customer": {
    "email": "mahnnooranwar191@gmail.com",
    "phone": "+9233503123560"
  },
  "orderItems": [
    {
      "product": "6a0d5da172241d4c9f9afe20c",
      "quantity": 2
    }
  ]
}
```

```
}
],
"shippingAddress": {
  "phoneNumber": "+9233503123560",
  "province": "Khyber Pakhtunkhwa",
  "city": "Chitral",
  "area": "malir",
  "fullAddress": "ce11 malir karachu",
  "landmark": "mega Mall",
  "deliveryInstructions": ""
},
"paymentMethod": "COD",
"couponCode": null,
"notes": "",
"saveToAddressBook": true
}
```

3. Response Structure (200 OK)

```
{
  "order": {
    "_id": "6a887c933aad01f4f71c4d83",
    "user": null,
    "customer": {
      "email": "mahnnooranwar191@gmail.com",
      "phone": "+9233503123560"
    },
    "orderItems": [
      {
        "product": "6a0d5da172241d4c9fafe20c",
        "name": "Hair Oil",
        "quantity": 2,
        "price": 1320,
        "originalPrice": 2200,
        "discountAmount": 1760,
        "shippingPrice": 250,
        "taxRate": 0,
        "subtotal": 2640
      }
    ],
    "shippingAddress": {
      "phoneNumber": "+9233503123560",
      "province": "Khyber Pakhtunkhwa",
```

```
"city": "Chitral",
"area": "malir",
"fullAddress": "ce11 malir karachu",
"lat": null,
"lon": null,
"landmark": "mega Mall",
"deliveryInstructions": ""
},
"paymentMethod": "COD",
"paymentStatus": "pending",
"transactionId": null,
"paymentDetails": null,
"orderStatus": "pending",
"itemsPrice": 2640,
"shippingPrice": 250,
"taxPrice": 0,
"discountAmount": 0,
"totalPrice": 2890,
"paidAt": null,
"notes": "",
"coupon": null,
"courierName": "Leopards Courier",
"trackingNumber": "LEO123456789",
"createdAt": "2026-08-21T16:28:03.754Z",
"updatedAt": "2026-08-21T16:28:03.754Z"
},
"coupon": null,
"totalPrice": 2890
}
```

SECTION 3: Flutter Integration Code (Dart)

1. Data Models

```
// --- Checkout Preview Models ---
class CheckoutPreviewResponse {
  final List<CheckoutItem> items;
  final PriceBreakdown priceBreakdown;
```

```

final dynamic coupon;
final String? couponError;

CheckoutPreviewResponse({
  required this.items,
  required this.priceBreakdown,
  this.coupon,
  this.couponError,
});

factory CheckoutPreviewResponse.fromJson(Map<String, dynamic> json) {
  return CheckoutPreviewResponse(
    items: (json['items'] as List)
      .map((item) => CheckoutItem.fromJson(item))
      .toList(),
    priceBreakdown: PriceBreakdown.fromJson(json['priceBreakdown']),
    coupon: json['coupon'],
    couponError: json['couponError'],
  );
}

class CheckoutItem {
  final String product;
  final String name;
  final String? image;
  final int quantity;
  final num price;
  final num originalPrice;
  final num shippingPrice;
  final num discountAmount;
  final num subtotal;

  CheckoutItem({
    required this.product,
    required this.name,
    this.image,
    required this.quantity,
    required this.price,
    required this.originalPrice,
    required this.shippingPrice,
    required this.discountAmount,
    required this.subtotal,
  });

  factory CheckoutItem.fromJson(Map<String, dynamic> json) {
    return CheckoutItem(
      product: json['product']?.toString() ?? "",

```

```

        name: json['name'] ?? "",
        image: json['image'],
        quantity: json['quantity'] ?? 0,
        price: json['price'] ?? 0,
        originalPrice: json['originalPrice'] ?? 0,
        shippingPrice: json['shippingPrice'] ?? 0,
        discountAmount: json['discountAmount'] ?? 0,
        subtotal: json['subtotal'] ?? 0,
    );
}
}

```

```

class PriceBreakdown {
    final num itemsPrice;
    final num productDiscount;
    final num shippingPrice;
    final num discountAmount;
    final num couponDiscount;
    final String? appliedCouponCode;
    final num totalPrice;
}

```

```

PriceBreakdown({
    required this.itemsPrice,
    required this.productDiscount,
    required this.shippingPrice,
    required this.discountAmount,
    required this.couponDiscount,
    this.appliedCouponCode,
    required this.totalPrice,
});

```

```

factory PriceBreakdown.fromJson(Map<String, dynamic> json) {
    return PriceBreakdown(
        itemsPrice: json['itemsPrice'] ?? 0,
        productDiscount: json['productDiscount'] ?? 0,
        shippingPrice: json['shippingPrice'] ?? 0,
        discountAmount: json['discountAmount'] ?? 0,
        couponDiscount: json['couponDiscount'] ?? 0,
        appliedCouponCode: json['appliedCouponCode'],
        totalPrice: json['totalPrice'] ?? 0,
    );
}
}

```

```

// --- Place Order Models ---
class CustomerInput {
    final String email;
    final String phone;
}

```

```
CustomerInput({required this.email, required this.phone});
```

```
Map<String, dynamic> toJson() => {  
  'email': email,  
  'phone': phone,  
};  
}
```

```
class ShippingAddressInput {  
  final String phoneNumber;  
  final String province;  
  final String city;  
  final String area;  
  final String fullAddress;  
  final String? landmark;  
  final String? deliveryInstructions;
```

```
  ShippingAddressInput({  
    required this.phoneNumber,  
    required this.province,  
    required this.city,  
    required this.area,  
    required this.fullAddress,  
    this.landmark,  
    this.deliveryInstructions,  
  });
```

```
  Map<String, dynamic> toJson() => {  
    'phoneNumber': phoneNumber,  
    'province': province,  
    'city': city,  
    'area': area,  
    'fullAddress': fullAddress,  
    'landmark': landmark ?? "",  
    'deliveryInstructions': deliveryInstructions ?? "",  
  };  
}
```

```
class PlaceOrderRequest {  
  final String? userId;  
  final CustomerInput customer;  
  final List<Map<String, dynamic>> orderItems;  
  final ShippingAddressInput shippingAddress;  
  final String paymentMethod;  
  final String? couponCode;  
  final String? notes;  
  final bool saveToAddressBook;
```

```

PlaceOrderRequest({
    this.userId,
    required this.customer,
    required this.orderItems,
    required this.shippingAddress,
    this.paymentMethod = 'COD',
    this.couponCode,
    this.notes = "",
    this.saveToAddressBook = false,
});

```

```

Map<String, dynamic> toJson() => {
    'userId': userId,
    'customer': customer.toJson(),
    'orderItems': orderItems,
    'shippingAddress': shippingAddress.toJson(),
    'paymentMethod': paymentMethod,
    'couponCode': couponCode,
    'notes': notes,
    'saveToAddressBook': saveToAddressBook,
};
}

```

```

class PlaceOrderResponse {
    final OrderData order;
    final dynamic coupon;
    final num totalPrice;

```

```

PlaceOrderResponse({
    required this.order,
    this.coupon,
    required this.totalPrice,
});

```

```

factory PlaceOrderResponse.fromJson(Map<String, dynamic> json) {
    return PlaceOrderResponse(
        order: OrderData.fromJson(json['order']),
        coupon: json['coupon'],
        totalPrice: json['totalPrice'] ?? 0,
    );
}

```

```

class OrderData {
    final String id;
    final String? trackingNumber;
    final String orderStatus;

```

```

final num totalPrice;

OrderData({
  required this.id,
  this.trackingNumber,
  required this.orderStatus,
  required this.totalPrice,
});

factory OrderData.fromJson(Map<String, dynamic> json) {
  return OrderData(
    id: json['_id'] ?? "",
    trackingNumber: json['trackingNumber'],
    orderStatus: json['orderStatus'] ?? 'pending',
    totalPrice: json['totalPrice'] ?? 0,
  );
}

```

2. API Service Implementation

```

import 'dart:convert';
import 'package:http/http.dart' as http;

class ApiService {
  static const String baseUrl = "https://your-domain.com"; // Replace with actual base URL

  // 1. Checkout Preview Call
  Future<CheckoutPreviewResponse> getCheckoutPreview({
    required List<Map<String, dynamic>> items,
    String? couponCode,
  }) async {
    final url = Uri.parse("$baseUrl/ogaglow/orders/checkout/preview");

    final body = {
      "items": items,
      if (couponCode != null && couponCode.isNotEmpty) "couponCode": couponCode,
    };

    final response = await http.post(
      url,
      headers: {
        'Content-Type': 'application/json',

```

```

    'Accept': 'application/json',
  },
  body: jsonEncode(body),
);

if (response.statusCode == 200) {
  return CheckoutPreviewResponse.fromJson(jsonDecode(response.body));
} else {
  final errorResponse = jsonDecode(response.body);
  throw Exception(errorResponse['message'] ?? 'Failed to calculate checkout preview');
}
}

// 2. Place Order Call
Future<PlaceOrderResponse> placeOrder(PlaceOrderRequest orderPayload) async {
  final url = Uri.parse('$baseUrl/ogaglow/orders/placeOrder');

  final response = await http.post(
    url,
    headers: {
      'Content-Type': 'application/json',
      'Accept': 'application/json',
    },
    body: jsonEncode(orderPayload.toJson()),
  );

  if (response.statusCode == 200 || response.statusCode == 201) {
    return PlaceOrderResponse.fromJson(jsonDecode(response.body));
  } else {
    final errorResponse = jsonDecode(response.body);
    throw Exception(errorResponse['message'] ?? 'Order placement failed');
  }
}
}

```

3. Key Notes for Flutter Developer

- **Automatic Backend Geocoding:** Backend multi-layered fallback geocoding (Landmark > Area > Address > City) run karta hai. Mobile app se coordinates (`lat`/`lon`) bhejne ki zaroorat nahi hai.
- **Guest vs Logged-In User:** Unauthenticated / Guest user ke liye `userId: null` pass karein. Server automatically user account handle/create karega.

- **Couriers & Tracking:** Order placement par auto Leopards Courier booking create hoti hai aur response mein `trackingNumber` return hota hai.
- **Debounce Preview Request:** UI mein quantity toggle ya coupon input par frequent preview calls ke liye debounce technique use karein.